CHM-110 | WEEK 10 EXAM CHEAT SHEET

Each question: full problem statement ⟶ how to recognize it ⟶ dimensional-analysis shortcut ⟶ answer.

01 Combustion gas stoichiometry — 23.8 L H_2O

PROBLEM: How many liters of gaseous H_2O are produced at 600. K and 850. Torr when 6.29 g of butane (C_4H_{10}) burn? $2\,C_4H_{10} + 13\,O_2 \longrightarrow 8\,CO_2 + 10\,H_2O$.

IDENTIFY: Grams of a reactant → liters of a different gas; pressure and temperature given (not STP).

EXAM SHORTCUT (UNITS):

$$6.29\,\text{g}\,C_4H_{10} \times \frac{1\,\text{mol}\,C_4H_{10}}{58.12\,\text{g}\,C_4H_{10}} \times \frac{10\,\text{mol}\,H_2O}{2\,\text{mol}\,C_4H_{10}}$$

$$V = 0.5411\,\text{mol}\,H_2O \times \frac{0.08206\,\text{L atm}}{\text{mol K}} \times \frac{600.\,\text{K}}{850.\,\text{Torr} \times \frac{1\,\text{atm}}{760\,\text{Torr}}}$$

02 Stoichiometry at STP — 28.8 L H_2

PROBLEM: How many liters of H_2 gas are produced when 59.2 g of Na react with excess water at STP? $2\,Na + 2\,H_2O \longrightarrow H_2 + 2\,NaOH$.

IDENTIFY: Grams of a substance → gas liters; **STP** specified. Use 22.4 L/mol.

EXAM SHORTCUT (UNITS):

$$59.2\,\text{g Na} \times \frac{1\,\text{mol Na}}{22.99\,\text{g Na}} \times \frac{1\,\text{mol}\,H_2}{2\,\text{mol Na}}$$

$$\times \frac{22.4\,\text{L}\,H_2}{1\,\text{mol}\,H_2} = 28.8\,\text{L}\,H_2$$

03 Sign of reaction enthalpy — Exothermic

PROBLEM: If $\Delta H = -73.54\,\text{kJ}$ for a chemical reaction, is it unable to occur, exothermic, or endothermic?

IDENTIFY: Only the sign of ΔH is needed; this is conceptual.

EXAM SHORTCUT (UNITS):

$$\Delta H = -73.54\,\text{kJ} < 0 \;\Rightarrow\; \text{heat released} \;\Rightarrow\; \text{exothermic}$$

$$\Delta H < 0:\ \text{exothermic}; \qquad \Delta H > 0:\ \text{endothermic}$$

No numerical calculation. Heat released by the system = negative ΔH.

04 Calorimetry: find mass — 2.23 kg Au

PROBLEM: How many kilograms of gold absorb 5310 J while warming from 20.0 °C to 38.3 °C? Specific heat of gold = 0.130 J/(g °C).

IDENTIFY: Energy, initial/final temperature and specific heat given; mass unknown.

EXAM SHORTCUT (UNITS):

$$q = mc(T_f - T_i) \quad\Longrightarrow\quad m = \frac{q}{c(T_f - T_i)}$$

$$\frac{5310\,\text{J}}{\left(0.130\,\frac{\text{J}}{\text{g}\cdot^\circ\text{C}}\right)(38.3 - 20.0)\,^\circ\text{C}} \times \frac{1\,\text{kg}}{1000\,\text{g}}$$

05 Molar enthalpy of solution — +25.7 kJ/mol

PROBLEM: When 0.873 g of ammonium nitrate (NH_4NO_3) dissolves, water cools; $\Delta H = +0.280\,\text{kJ}$. Which description of $\Delta H^\circ_{\text{soln}}$ and heat flow is correct?

IDENTIFY: Mass and kJ supplied; options give kJ/mol and absorbed/released.

EXAM SHORTCUT (UNITS):

$$\Delta H^\circ_{\text{soln}} = \frac{0.280\,\text{kJ}}{0.873\,\text{g}\,NH_4NO_3} \times \frac{80.052\,\text{g}\,NH_4NO_3}{1\,\text{mol}\,NH_4NO_3}$$

$$= +25.7\,\text{kJ/mol} \quad\Rightarrow\quad \text{endothermic (heat absorbed)}$$

KEY Recognize the conversion path — Read units first

PROBLEM: For any gas, heat, or reaction problem, identify the supplied and requested units before selecting a formula.

IDENTIFY: Different chemicals require a reaction coefficient ratio; the same gas with P, V, n, T requires $PV = nRT$.

EXAM SHORTCUT (UNITS):

$$\text{g A} \;\rightarrow\; \text{mol A} \xrightarrow{\text{coefficient ratio}} \text{mol B} \;\rightarrow\; \text{g or L B}$$

STP: 22.4 L/mol; otherwise: $PV = nRT$

Round only at the end; keep units on each conversion factor.

06 Matching energy terms and diagrams

1, 1, 1, 2, 3, 2, 1, 1

PROBLEM: Match each description with (1) endothermic, (2) exothermic, or (3) specific heat capacity. The graph shows reactants near 50 kJ and products near 100 kJ.

(a) A reaction in which energy is absorbed.
(b) Reaction-pathway graph: products higher than reactants.
(c) A reaction in which energy is on the reactant side.
(d) A reaction with energy on the product side.
(e) Energy required to raise 1 g by 1 °C.
(f) A reaction that feels warm.
(g) Products have more energy than reactants.
(h) A reaction with positive ΔH.

IDENTIFY: Heat absorbed or products higher = endothermic; heat released or products lower = exothermic.

$$\Delta H = H_{\text{products}} - H_{\text{reactants}} \quad (100 - 50 = +50\,\text{kJ}: \ \text{endothermic})$$

07 Gas liters to solid grams

173 g

PROBLEM: 25.0 L of CO_2 form when $Pb(CO_3)_2$ decomposes at 5.30 atm and 1250 °C. How many grams decomposed? $Pb(CO_3)_2 \longrightarrow 2\,CO_2 + PbO_2$.

IDENTIFY: P, V, T of gaseous product given; mass of a different solid reactant requested.

EXAM SHORTCUT (UNITS):

$$n_{CO_2} = \frac{(5.30\,\text{atm})(25.0\,\text{L})}{\left(0.08206\,\frac{\text{L atm}}{\text{mol K}}\right)(1250 + 273.15)\,\text{K}}$$

$$1.060\,\text{mol}\,CO_2 \times \frac{1\,\text{mol}\,Pb(CO_3)_2}{2\,\text{mol}\,CO_2} \times \frac{327.22\,\text{g}\,Pb(CO_3)_2}{1\,\text{mol}\,Pb(CO_3)_2}$$

08 Sign of reaction enthalpy

Exothermic

PROBLEM: If $\Delta H = -54.74\,\text{kJ}$ for a chemical reaction, does the reaction not occur, release heat, or absorb heat?

IDENTIFY: Negative ΔH; concept question identical in type to Q03.

EXAM SHORTCUT (UNITS):

$$\Delta H = -54.74\,\text{kJ} < 0 \quad \Rightarrow \quad \text{exothermic}$$

Negative ΔH $\Rightarrow$ heat is released to surroundings

No numerical calculation.

09 Ideal Gas Law: pressure

2.28 atm

PROBLEM: What is the pressure in a 5.80 L container holding 0.545 mol of O_2 at 22.0 °C?

IDENTIFY: Moles, volume, temperature known; pressure unknown; no reaction.

EXAM SHORTCUT (UNITS):

$$P = \frac{nRT}{V}$$

$$P = \frac{(0.545\,\text{mol})\left(0.08206\,\frac{\text{L atm}}{\text{mol K}}\right)(22.0 + 273.15)\,\text{K}}{5.80\,\text{L}}$$

10 Thermochemistry: kJ to grams

357 g

PROBLEM: How many grams of $HC_2H_3O_2$ are consumed when 936 kJ are absorbed? $MgCO_3 + 2\,HC_2H_3O_2 \longrightarrow Mg(C_2H_3O_2)_2 + H_2CO_3$; $\Delta H = +315\,\text{kJ}$.

IDENTIFY: Energy given, ΔH written for balanced reaction, grams requested.

EXAM SHORTCUT (UNITS):

$$936\,\text{kJ} \times \frac{2\,\text{mol}\,HC_2H_3O_2}{315\,\text{kJ}} \times \frac{60.052\,\text{g}\,HC_2H_3O_2}{1\,\text{mol}\,HC_2H_3O_2} = 357\,\text{g}$$

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11 Molecules to gas liters — 395 L

PROBLEM: What is the volume of 7.32×10^{24} molecules of gas at 768 Torr and 400. K?

IDENTIFY: Molecules (not mol) given; pressure and temperature supplied; volume requested.

EXAM SHORTCUT (UNITS):

$$7.32 \times 10^{24}\,\text{molecules} \times \frac{1\,\text{mol}}{6.022 \times 10^{23}\,\text{molecules}} = 12.155\,\text{mol}$$

$$V = 12.155\,\text{mol} \times \frac{0.08206\,\text{L atm}}{\text{mol K}} \times \frac{400.\,\text{K}}{768\,\text{Torr} \times \frac{1\,\text{atm}}{760\,\text{Torr}}}$$

12 Ideal Gas Law: find grams — 6.46 g

PROBLEM: How many grams of dinitrogen trioxide (N_2O_3) are in a 530. mL container at 28.0 °C and 3.96 atm?

IDENTIFY: Given P, V, T for a single gas; mass requested; no reaction. Convert mL to L.

EXAM SHORTCUT (UNITS):

$$530.\,\text{mL} \times \frac{1\,\text{L}}{1000\,\text{mL}} = 0.530\,\text{L}$$

$$n = \frac{(3.96\,\text{atm})(0.530\,\text{L})}{\left(0.08206\,\frac{\text{L atm}}{\text{mol K}}\right)(28.0 + 273.15)\,\text{K}}$$

$$0.08493\,\text{mol}\,N_2O_3 \times \frac{76.02\,\text{g}\,N_2O_3}{1\,\text{mol}\,N_2O_3}$$

13 STP gas liters to kJ — 114 kJ released

PROBLEM: How many kJ are released when 83.2 L of H_2 are consumed at STP? Supplied equation: $N_2 + 3\,H_2 \longrightarrow NH_3 + 92.0$ kJ.

IDENTIFY: Gas liters at STP, kJ on product side; convert liters → mol → kJ.

EXAM SHORTCUT (UNITS):

$$83.2\,\text{L}\,H_2 \times \frac{1\,\text{mol}\,H_2}{22.4\,\text{L}\,H_2} \times \frac{92.0\,\text{kJ}}{3\,\text{mol}\,H_2}$$

$$= 114\,\text{kJ released}$$

Quiz equation is missing the 2 before NH_3; the intended $3\,\text{mol}\,H_2 : 92.0\,\text{kJ}$ remains unchanged.

14 Calorimetry: heat released — 5500 cal released

PROBLEM: How many calories are absorbed or released when 100. g of H_2O cool from 80. °C to 25 °C? Water's specific heat = 1 cal/(g °C).

IDENTIFY: Mass, specific heat, two temperatures; solve for heat q. Cooling means negative q.

EXAM SHORTCUT (UNITS):

$$q = mc(T_f - T_i)$$

$$q = (100.\,\text{g})\left(1\,\frac{\text{cal}}{\text{g} \cdot {}^\circ\text{C}}\right)(25 - 80)\,{}^\circ\text{C}$$

$$= -5500\,\text{cal} \quad \Rightarrow \quad 5500\,\text{cal released}$$

REFERENCE Exam formula and factor-label key — Use exact units

IDENTIFY: Gas formulas and conversions

$$PV = nRT; \qquad P = \frac{nRT}{V}; \quad V = \frac{nRT}{P}; \quad n = \frac{PV}{RT}$$

$R = 0.08206\,\text{L atm/(mol K)}$; $T(K) = T(^\circ C) + 273.15$; $1\,\text{atm} = 760\,\text{Torr}$.

$1\,\text{mol gas} = 22.4\,\text{L}$ at STP; $1\,\text{mol} = 6.022 \times 10^{23}$ particles.

IDENTIFY: Heat, stoichiometry, and unit rules

$$q = mc\Delta T; \quad \Delta T = T_f - T_i; \quad m = nM; \quad n = \frac{m}{M}$$

Reaction coefficients = molar ratios; ΔH per reaction gives kJ-to-mol ratios.

$1\,\text{kg} = 1000\,\text{g}$; $1\,\text{L} = 1000\,\text{mL}$; positive ΔH absorbs heat.